

AUDITING GRADIENT RETENTION UNDER HIDDEN CONDITIONS: EXACT CANCELLATION, OBJECTIVE-DEPENDENT ALIGNMENT, AND THE LIMITS OF KAPPA

Anonymous authors

Paper under double-blind review

ABSTRACT

Training data is often a mixture of hidden conditions: annotator groups with conflicting preferences, clients with different objectives, teammates with unobserved roles. When a single model is trained on the mixture, the update is an average of condition-specific gradients, and part of the signal can cancel before it reaches the weights. We study when this cancellation is exact, when it is partial, and how it must be measured. On a mirror bandit we prove that elementwise, parameter-independent weight fields cancel exactly under the symmetric hidden mixture, and we derive an exact closed form for the soft-Q field. On a K -way matching bandit we show that uniform mixtures cancel for every K and that direction dependence first appears once the condition simplex has dimension at least two; the entropy and mixture channels of the soft-Q field can also interfere. A measurement audit shows that deterministic rollouts can turn the retention score into a weight ratio, and that episode-noise readouts can differ by an order of magnitude from the structural score. Refreshed measurements under one bounded definition across a hidden matching task, Overcooked, and a card-game environment show that value fields align with hidden conditions (structurally up to 0.999) while policy-gradient fields remain near-orthogonal (around 0.5); the effect is real but about twice, not thirty times. Finally, retention does not predict task performance: a TD field with the lowest measured retention achieves the best return in a switching task. Condition-gradient auditing must separate cancellation, alignment, and noise before any optimizer-level intervention.

1 INTRODUCTION

Supervised and reinforcement learning systems are routinely trained on data that mixes heterogeneous subpopulations. The subpopulation label is frequently missing: a preference dataset pools annotators with conflicting tastes, a federated client population encodes different task distributions, a multi-agent episode hides which role the partner plays. The model sees one input distribution, while the label or reward is generated by a condition that is not present in the input. Training then optimizes the mixture objective, and the update direction is the mixture of condition-specific gradients,

$$\bar{g} = \sum_i p_i g_i, \quad (1)$$

where $g_i(\theta) = \nabla_{\theta} L_i(\theta)$ is the population gradient of the condition- i objective under a fixed rollout distribution.

A mixture can erase signal. The extreme case is cancellation: if two conditions demand opposite behavior on the same input, their gradients can point in opposite directions and the average update is zero even though each condition carries a strong, learnable signal. Practitioners observe this as training that stalls without any visible loss anomaly. This paper asks how to audit that failure

precisely: when cancellation is exact, when it is partial, how it depends on the objective, and what the resulting measurement does and does not tell us.

A companion phenomenon was reported previously: hidden relational variables can contract policy-gradient updates while value-based updates appear to survive. That report left the mechanism open. Our starting point is that the mechanism cannot be recovered from algorithm-family labels, because a gradient field is a property of the *objective evaluated on a fixed policy and data*, not of the optimizer. We therefore fix the measurement basis: same network, same rollout distribution, same condition definition, and change only the scalar objective whose gradient defines the field.

Contributions.

1. **Exact cancellation criteria.** For the mirror bandit, any elementwise, parameter-independent weight field satisfies $\mathbb{E}[g_B] = -\mathbb{E}[g_A]$, hence zero shared gradient under the symmetric mixture. The same normalization yields exact cancellation for K -way matching mixtures at every K . We state the boundaries: these are statements about expected gradients on the stated model, not about arbitrary neural environments.
2. **Objective-dependent alignment with exact formulas.** We give a closed form for the soft-Q field, $\kappa_{\text{mix}} = \Delta^2/(\Delta^2 + 4r^2)$ with $\Delta = \alpha \log(\rho/(1-\rho))$, verified against autograd to machine precision, and we isolate the advantage-weighted field with a differentiable baseline as the only policy-gradient-style field that survives the symmetric mixture, through an implicit even coupling.
3. **K -way geometry.** Uniform mixtures cancel for $K = 2, 3, 4$; direction dependence at fixed distance from uniform appears at $K \geq 3$; and entropy and mixture channels interfere with a computable minimum. These are properties of the condition simplex, not of a non-abelian group.
4. **Measurement audit.** Deterministic rollouts reduce the retention score to a single-action weight ratio on the mirror model; a measured example gives 0.633 where the correct protocol gives 0.001. Episode-noise scores can be an order of magnitude below structural scores, so both must be reported.
5. **Refreshed real-environment evidence.** Under one bounded structural definition, value fields align strongly with hidden conditions (Overcooked dynamic 0.999, 510K 0.95–0.98) while policy-gradient fields remain near-orthogonal (about 0.51–0.53). The gap is about $1.9\times$, not the $30\text{--}65\times$ implied by noise-dominated readouts.
6. **A boundary result.** Retention is not a performance score. In a switching task, the TD field has the lowest measured retention and the highest return, while a high-retention soft-Q field performs worse. We explain the decoupling by variable mismatch and use it to scope the diagnostic.

2 RELATED WORK

Gradient conflict and multi-task optimization. PCGrad (Yu et al., 2020), CAGrad (Liu et al., 2021), MGDA (Sener & Koltun, 2018), GradNorm (Chen et al., 2018), GradDrop (Chen et al., 2020), and later gradient-surgery methods treat conflict as an observed quantity to be projected, reweighted, or scheduled away. We study the complementary question: when conflict is *structural*, i.e. forced by hidden-condition symmetry before any optimizer intervenes. Our audit is a diagnostic, not another surgery rule, and we do not claim to replace those methods.

Hidden context and preference aggregation. Distributional preference learning (Siththaranjan et al., 2024) shows that averaging unobserved annotator groups distorts a reward model; heterogeneous-preference alignment work studies minority under-representation and equitable aggregation (Chakraborty et al., 2024). We work in gradient space rather than risk or reward space and provide exact cancellation conditions plus objective-specific survival rates; we do not propose a new alignment algorithm.

Symmetry, averaging, and invariance. Averaging over a group acts as a projection onto invariant components (a Reynolds operator), and the resulting decomposition is classical. Our use of it

Table 1: Measured fields. Every entry is a gradient with respect to the same policy parameters on the same rollouts; only the scalar objective changes.

Field	Objective or weight	Note
reinforce	$\mathbb{E}_a[\nabla(-\log \pi(a) Q_r(a))]$	parameter-free weight
expq	$\nabla(-\sum_a \pi(a) Q_r(a))$	full objective
softq	$\nabla \sum_a \pi(a)(\alpha \log \pi(a) - Q_r(a))$	detached Q
awr	$\mathbb{E}_a[\nabla(-\log \pi(a) e^{(Q_r(a)-V)/\tau})]$	differentiable V
softmaxq	$\mathbb{E}_a[\nabla(-\log \pi(a) \text{softmax}(Q_r/\tau)_a)]$	Q -dependent weight

is instrumental: we apply it to condition-conditioned gradient fields and derive measurement consequences. We explicitly do not claim group averaging as a contribution.

The companion report. A prior study (citation withheld for double-blind review) introduced the retention ratio and reported estimator dependence and an apparent policy-gradient/value reversal. That report is a phenomenon study. The present work replaces the algorithm comparison with a common-basis field measurement, derives the closed forms, and corrects two measurement practices on which the earlier scores depended.

3 PROBLEM SETUP AND MEASUREMENT

3.1 HIDDEN-CONDITION GRADIENT FIELDS

Let i index a hidden condition with mixture weights p_i , and let $g_i = \nabla_{\theta} L_i$ denote the population condition gradient of a scalar objective L_i under a fixed rollout distribution. We report the bounded structural retention

$$\kappa_{\text{mix}} = \frac{\|\bar{g}\|^2}{\sum_i p_i \|g_i\|^2}, \quad 0 \leq \kappa_{\text{mix}} \leq 1, \quad (2)$$

where the upper bound follows from convexity. We also report the energy decomposition

$$E_{\text{shared}} = \|\bar{g}\|^2, \quad E_{\text{contrast}} = \sum_i p_i \|g_i - \bar{g}\|^2, \quad E_{\text{mix}} = \sum_i p_i \|g_i\|^2. \quad (3)$$

The historical uniform-reference normalization $\|\bar{g}\|^2 / \text{mean}_i \|g_i\|^2$ is not bounded for non-uniform p and is used only to read legacy result files.

3.2 FIELDS UNDER STUDY

All fields are gradients with respect to the same policy parameters on the same data (Table 1).

The distinction between a differentiable and a detached baseline matters: only the former changes the parity structure of the field, so we refer to our awr as an advantage-weighted field with a differentiable baseline rather than as standard AWR.

3.3 MEASUREMENT PROTOCOL

Three rules follow from the audit in Section 5.2 and are used throughout. (i) Rollouts are stochastic and policy-weighted; deterministic (argmax) rollouts collapse the score into a weight ratio on the mirror model. (ii) Conditions are forced per rollout when the environment allows it, so the measured contrast is between clean condition assignments rather than between random mixes. (iii) When two fields are compared, both are computed from the *same* episode batch. We report structural scores from condition means and, separately, the one-episode noise diagnostic

$$\kappa_{\text{ep}} = \frac{E_{\text{shared}}}{E_{\text{mix}} + \sum_i p_i \sigma_i^2}, \quad (4)$$

where σ_i^2 is the within-condition episode-gradient variance. A finite-sample mean calibration, $E_{\text{shared}} / (E_{\text{mix}} + \sum_i p_i \sigma_i^2 / n_i)$, is reported as a surrogate and is not a high-probability guarantee.

4 THEORY

4.1 EXACT CANCELLATION

Proposition 1 (Mirror cancellation). *In the two-action mirror bandit with $Q_A = (r, -r)$ and $Q_B = (-r, r)$, any field whose weight is elementwise and independent of the policy parameters satisfies $\mathbb{E}[g_B] = -\mathbb{E}[g_A]$. Under the symmetric mixture $p = (1/2, 1/2)$, $\mathbb{E}[\bar{g}] = 0$ and $\kappa_{\text{mix}} = 0$ exactly.*

Verification. Closed form and autograd agree at the 10^{-32} level for the stored configurations, covering reinforce, expq, awr without baseline, and softmaxq (data/kappa/toy-fields/ol-closed-forms.json).

Proposition 2 (K -way normalization). *In the K -way matching bandit with $Q_i(b) = 2R\delta(i, b)$ and a uniform condition mixture, $\sum_i g_i = 0$ for every K ; the cancellation is a normalization identity, not a group-theoretic accident. Verified for $K = 2, 3, 4$ with $\kappa_{\text{mix}} < 1.8 \times 10^{-32}$.*

The boundary is explicit: the result concerns expected gradients under the matching construction. Real environments are not exact mirrors, and the measured structural scores there are around 0.5, i.e. near-orthogonal rather than cancelling.

4.2 CLOSED FORM FOR THE SOFT-Q FIELD

Proposition 3 (Soft-Q retention). *In the mirror bandit, let $\rho = \pi(\text{action } 0)$, $q = 1 - \rho$, and $\Delta = \alpha \log(\rho/q)$. The soft- Q field satisfies*

$$c_A = \rho q(\Delta - 2r), \quad c_B = \rho q(\Delta + 2r), \quad g_i = c_i u, \quad \kappa_{\text{mix}} = \frac{\Delta^2}{\Delta^2 + 4r^2}. \quad (5)$$

Verification. Closed form versus autograd of the explicit expected loss agrees to 5.0×10^{-16} ; Monte Carlo anchors agree within sampling error; the stored α -grid values match to the archived precision (one archived point differs by 0.001 under a documented old rounding convention; verify-closed-forms-ol.py). Limiting cases: $\alpha \rightarrow 0$ degenerates to the cancelling expected- Q field, while a non-uniform policy or a larger entropy weight increases retention. This is a fixed-policy identity; it does not describe a training trajectory.

The advantage-weighted field with a differentiable baseline is the only policy-gradient-style field that survives the symmetric mixture: the baseline enters the weight through $V = \sum_a \pi(a)Q(a)$ and creates an implicit even coupling. Its closed form appears in Appendix A; at uniform policies it also cancels for every temperature, and for non-uniform policies retention is non-monotone in τ .

4.3 K -WAY GEOMETRY AND CHANNEL INTERFERENCE

Proposition 4 (Direction dependence). *For the K -way matching field, $g_i[b] = -2R\pi_b(\delta(i, b) - \pi_i)$ and $\bar{g}[b] = -2R\pi_b(p_b - \langle p, \pi \rangle)$, where the second term is the projection of the mixture onto the policy-orthogonal complement. When the condition simplex has dimension at least two ($K \geq 3$), retention at fixed L_1 distance from the uniform mixture varies with direction.*

Measured spread at L_1 distance 0.45 (200 directions): $K = 3$: $\kappa_{\text{mix}} \in [0.071, 0.219]$ ($3.10\times$); $K = 4$: $[0.051, 0.210]$ ($4.08\times$). This is simplex geometry; any claim that the effect is specific to S_3 or to non-abelian symmetry is withdrawn. Direction dependence is absent for $K = 2$ because the mixture simplex is one-dimensional.

Proposition 5 (Channel interference). *The soft- Q field is the sum of an entropy channel and a condition-mixture channel in the same weighted tangent space. When these two directions have positive weighted inner product, retention is non-monotone in α : the numerator cancels before the ratio recovers.*

Measured minima: $\kappa_{\text{mix}} = 7.8 \times 10^{-4}$ at $\alpha = 1.78$ for $K = 2$ with $p = (0.8, 0.2)$, and 1.1×10^{-3} at $\alpha = 1.41$ for $K = 3$ with $p = (0.7, 0.2, 0.1)$. The numerator cancellation point and the ratio minimum are distinct and must be reported separately.

Table 2: Overcooked structural retention (three seeds). The value entry is the diagnostic field $-\sum_t V(s_t)$; it is not a TD residual.

Field	static	dynamic
reinforce	0.515 ± 0.021	0.529 ± 0.034
awr	0.516 ± 0.023	0.573 ± 0.015
value ($-\sum V$)	0.568 ± 0.027	0.999 ± 0.000

A dynamical reading of this interference was proposed previously and is withdrawn here. Under the implemented objective $L = \sum_a \pi(a)(\alpha \log \pi(a) - Q(a))$, gradient descent on the symmetric mixture minimizes $-\alpha H(\pi)$ and therefore moves toward higher entropy, not toward a peaked policy. Static channel interference survives as a statement about the field, not about an attractor.

5 EXPERIMENTS

All numbers are produced by the canonical implementation and collected in `notes/canonical_metric_results.md`; raw files are listed in Appendix B.

5.1 EXACT BANDIT: CANCELLATION AND CLOSED FORMS

On the hidden mirror bandit with a fixed near-uniform policy, `expq` cancels exactly ($\kappa_{\text{mix}} = 0$, $E_{\text{shared}} = 0$, with $E_{\text{mix}} = 1.81$) and `reinforce` is indistinguishable from zero within sampling noise (0.0011 ± 0.0013), while the soft-Q field retains a small shared component (0.016 at $\alpha = 1$) that grows monotonically with α and follows Proposition 3. On the revealed policy all five fields collapse to the same value (0.412 – 0.420), the expected degeneracy of a policy that already solves the matching task. The contrast between 0 and 0.41 on the same parameters and data, with only condition visibility changed, is the cleanest demonstration that hidden conditions can erase a field.

5.2 PROTOCOL AUDIT: DETERMINISTIC ROLLOUTS INVENT A FIELD AXIS

On the hidden mirror bandit, deterministic (`argmax`) rollouts visit a single action per condition, and every field reduces to the ratio of weights at that action. Measured artifacts: `softmaxq` reports 0.633 under deterministic rollouts and 0.0011 under the correct stochastic protocol; `awr` reports 0.551 versus 0.028. `expq` remains exactly zero under both protocols, and the revealed-policy values are protocol-robust (~ 0.396 both ways). The audit rule is direct: an apparent field axis measured with `argmax` rollouts on a degenerate model should be treated as a weight ratio until recomputed with policy-weighted sampling.

5.3 REAL ENVIRONMENTS UNDER THE CANONICAL METRIC

Overcooked (switching-preserving protocol). With the same fresh policy, the same episode batch, and only the measured objective changed, structural retention separates the fields (Table 2). The value field’s condition-mean gradients nearly coincide on the dynamic model, while the policy-gradient fields are near-orthogonal. Under the start-partner protocol the same conclusion appears at the level of condition means (static 0.533, dynamic 0.502), and a memory intervention does not change it (m4 0.498, m8 0.505, m16 0.500): supplying memory does not restore alignment, a negative intervention result.

510K (paired protocol). Using checkpoints trained with hidden teammate visibility $p \in \{0, 0.5, 1\}$ and computing both fields from the same episode batch, per-step reward-weighted policy gradients retain 0.514/0.514/0.513 while value fields retain 0.980/0.974/0.948 (Table 3). The ordering is stable at every seed; the visible/hidden difference is small and we do not interpret it as a trend.

Table 3: 510K structural retention, paired protocol (six seeds per level). Both fields are computed from the same episode batch.

Visibility p	reinforce	value
0.0 (hidden)	0.514 ± 0.099	0.980 ± 0.011
0.5	0.514 ± 0.084	0.974 ± 0.018
1.0 (visible)	0.513 ± 0.092	0.948 ± 0.017

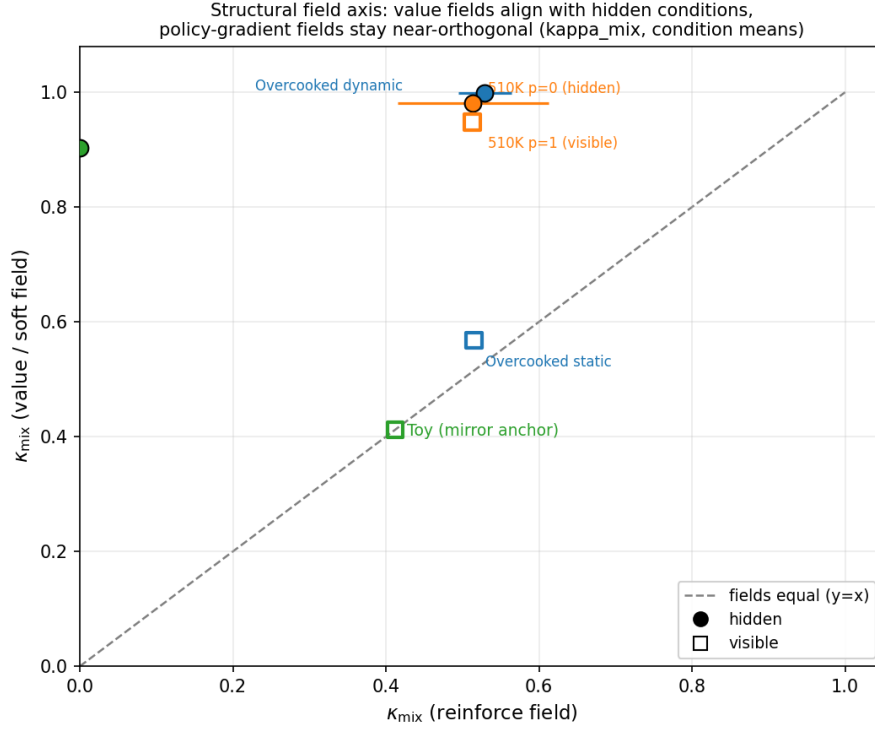


Figure 1: Structural field axis under κ_{mix} (condition means). Value fields sit above the diagonal; policy-gradient fields stay near 0.5 in the real environments. The mirror-bandit point is an exact anchor, not a real environment.

Reading. Across the two environments the value fields align with hidden conditions while policy-gradient fields sit near 0.5 (orthogonal). Expressed as a ratio this is about $1.9\times$. The previously reported $30\text{--}65\times$ separations came from comparing one-episode noise scores (κ_{ep} , where the value field’s within-condition variance is small and the policy-gradient field’s variance is large) rather than structural scores. We report both quantities, but only the structural one supports field-level statements.

5.4 CAPACITY, NOT OPTIMIZER

On an exact K -way training comparison with a shared policy versus a condition-aware policy (full-batch, deterministic, same initial logits), the shared model’s limitation is a mixture compromise (Table 4). With a uniform mixture and the linear objective, the shared model stays at average reward 0.667 while the conditional model reaches 1.99 ($\kappa_{\text{mix}} = 0$ at initialization, by cancellation). Under mild and strong asymmetric mixtures the shared model improves the average (1.20 and 1.60) but leaves the worst condition at 0.003 and 0.002, while the conditional model keeps the worst condition at 1.97 and 1.93. Turning on the entropy channel ($\alpha = 2$) trades average for worst-condition performance in the shared model (0.752 average, 0.517 worst), a Pareto movement rather than a

Table 4: Shared versus condition-aware capacity (exact full-batch training). Rewards are reported as average / worst condition.

Mixture	Model	avg. reward	worst reward	initial κ_{mix}
uniform	shared, $\alpha=0$	0.667	0.466	0.000
uniform	conditional, $\alpha=0$	1.992	1.992	—
mild	shared, $\alpha=0$	1.197	0.003	0.208
mild	conditional, $\alpha=0$	1.991	1.970	—
mild	shared, $\alpha=2$	0.752	0.517	0.445
strong	shared, $\alpha=0$	1.597	0.002	0.501
strong	conditional, $\alpha=0$	1.991	1.933	—

Table 5: Switching task: objectives trained from scratch under an identical budget. Post-switch reward is measured on steps 20–28 after the flip.

Arm	return	post-switch reward	final κ_{mix}
td	36.97 ± 0.06	0.75	0.06 ± 0.04
softq	28.62 ± 11.81	0.58	0.94 ± 0.09
reinforce	9.48 ± 5.13	0.18	0.43 ± 0.18
awr	5.03 ± 2.02	0.04	0.57 ± 0.08

fix. The intervention suggested by the audit is conditional information or capacity, not an arbitrary gradient rule.

5.5 RETENTION IS NOT A PERFORMANCE SCORE

In a switching hidden-matching task with four objectives trained from scratch under an identical budget (eight seeds), final returns and final structural retention are shown in Table 5. The arm with the lowest measured retention is the best performer. The explanation is variable mismatch: the task requires tracking a phase from reward history, which is orthogonal to the initial partner label that retention conditions on. The diagnostic answers “how much of the specified condition signal survives averaging,” not “will training succeed.” A low retention score must therefore be paired with a check that the condition is the variable the task actually relies on.

6 WHAT THE AUDIT DOES AND DOES NOT CLAIM

The retained statements are: (i) exact cancellation conditions on the mirror and matching models; (ii) an exact closed form for the soft-Q field and the differentiable-baseline exception; (iii) K -way direction dependence and channel interference as geometry; (iv) protocol and noise separation as methodology; (v) structural alignment of value fields versus near-orthogonality of policy-gradient fields in two environments; and (vi) decoupling from performance. We explicitly do not claim that value methods are universally better, that policy-gradient fields always vanish, that direction dependence is group-specific, that retention predicts return, or that memory or entropy interventions fix cancellation.

7 LIMITATIONS

The exact results are bandit-level. Real-environment claims rest on condition-mean gradients under one measurement protocol and on small seed counts (three seeds for the Overcooked field axis). The value field is the diagnostic $-\sum V$, not a TD residual; its alignment is a property of the fitted value function on the measured distribution. The 510K environment has weak relational drive and is used as a supporting measurement, not as the primary demonstration. No real-world heterogeneity dataset is included; a natural next step is a three-group data setting with a shared model and a

condition-aware routing baseline, reporting average and worst-group metrics. The finite-sample analysis reports a deterministic surrogate; a high-probability guarantee for the ratio estimator would require an explicit distributional assumption.

REPRODUCIBILITY STATEMENT

The metric implementation is `src/iigc/metrics/kappa.py`; the closed forms, protocol audit, K -way controls, Overcooked and 510K measurements, and the capacity comparison are scripted under `experiments/common_basis/`. Result files carry a `measurement_metadata` block recording the gradient definition, condition weights, rollout protocol, parameter space, and noise treatment. Unit tests cover the closed forms, cancellation, boundedness, and the metadata contract. No human subjects or private data are involved. Full provenance is listed in Appendix B.

AI USE STATEMENT

In this work, we used generative AI tools for drafting assistance and code editing suggestions. We have reviewed and verified all AI-assisted work: all closed-form results were checked against independent autograd and Monte Carlo computations, and all reported numbers are regenerated by the scripts listed in Appendix B. We take responsibility for the final content of this work.

ETHICS STATEMENT

This work studies optimization diagnostics for models trained on mixtures of hidden subpopulations. It does not involve human subjects or private data. The discussed failure mode (minority conditions being under-represented by averaged objectives) has fairness implications; we present the audit as a diagnostic and make no deployment recommendation.

REFERENCES

- Souradip Chakraborty, Jiahao Qiu, Hui Yuan, Alec Koppel, Furong Huang, Dinesh Manocha, Amrit Singh Bedi, and Mengdi Wang. Maxmin-rlhf: Alignment with diverse human preferences. In *International Conference on Machine Learning*, 2024.
- Zhao Chen, Vijay Badrinarayanan, Chen-Yu Lee, and Andrew Rabinovich. Gradnorm: Gradient normalization for adaptive loss balancing in deep multitask networks. In *International Conference on Machine Learning*, 2018.
- Zhao Chen, Jiquan Ngiam, Piyush Sapra, Qizhen Le, et al. Just pick a sign: Optimizing deep multitask models with gradient sign dropout. In *Advances in Neural Information Processing Systems*, 2020.
- Bo Liu, Xingchao Liu, Xiaojie Jin, Peter Stone, and Qiang Liu. Conflict-averse gradient descent for multi-task learning. In *Advances in Neural Information Processing Systems*, 2021.
- Ozan Sener and Vladlen Koltun. Multi-task learning as multi-objective optimization. In *Advances in Neural Information Processing Systems*, 2018.
- Anand Siththaranjan, Cassidy Laidlaw, and Dylan Hadfield-Menell. Distributional preference learning: Understanding and accounting for hidden context in rlhf. In *International Conference on Learning Representations*, 2024.
- Tianhe Yu, Saurabh Kumar, Abhishek Gupta, Sergey Levine, Karol Hausman, and Chelsea Finn. Gradient surgery for multi-task learning. In *Advances in Neural Information Processing Systems*, 2020.

A DIFFERENTIABLE-BASELINE CLOSED FORM

For the advantage-weighted field with $w = \exp((Q_r - V)/\tau)$ and a differentiable baseline $V = \sum_a \pi(a)Q_r(a)$, write $x = 2r/\tau$,

$$A_0 = e^{xq}, \quad A_1 = e^{-xp}, \quad M_A = p \log p A_0 + q \log q A_1, \quad (6)$$

$$B_0 = e^{-xq}, \quad B_1 = e^{xp}, \quad M_B = p \log p B_0 + q \log q B_1, \quad (7)$$

with $p = \pi(\text{action } 0)$, $q = 1 - p$. Then

$$c_A = pq(A_1 - A_0 + xM_A), \quad c_B = pq(B_1 - B_0 - xM_B). \quad (8)$$

At $p = 1/2$ the field cancels exactly for every τ ; otherwise retention has an interior maximum in τ . Closed form and autograd agree to 9.2×10^{-16} and Monte Carlo anchors agree within three standard errors (`verify_closed_forms_01.py`).

B NUMBER PROVENANCE

Table 6: Provenance of the main-text numbers.

Result	File or script
Cancellation, soft-Q and AWR closed forms	<code>data/kappa/toy_fields/01_closed_forms.json</code>
K -way cancellation, direction, interference	<code>data/kappa/toy_fields/kway_geometry.json</code>
Protocol audit (deterministic vs stochastic)	<code>data/kappa/toy_fields/det_vs_stoch.json</code>
Overcooked field axis	<code>data/kappa/server_tasks/results/oc_field_axis.json</code>
Overcooked switch and memory protocol	<code>data/kappa/server_tasks/results/oc_switch_kappa.json</code>
510K paired protocol	<code>data/kappa/server_tasks/results/510k_field_axis.json</code>
Shared vs conditional capacity	<code>data/kappa/compromise_training/results.json</code>
Switching-task decoupling	<code>data/kappa/o4_adaptive/aggregate.json</code>
Canonical metric implementation	<code>src/iigc/metrics/kappa.py</code>

C CORRECTION LOG

Superseded claims and their replacements are recorded in `notes/paper1_propositions_corrections.md`, including the withdrawn soft-Q dynamics claim, the parameter-space scope of the Fisher/KL variant, and the replacement of S_3 -specific language with K -way simplex geometry.